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University of Applied Sciences

Fraunhofer
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Master's Thesis

Multi-objective Optimisation of suction-cup and two finger grippers using todo: <Algorithims>

im Studiengang Automotive Systems
der FakultÃ¤t Fahrzeugtechnik
todo: <Semester>

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Zeitraum: todo: <Zeitraum>
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Eidesstattliche Erklärung

Hiermit versichere ich, die vorliegende Arbeit selbstständig und unter ausschließlicher Verwendung der angegebenen Literatur und Hilfsmittel erstellt zu haben.

Die Arbeit wurde bisher in gleicher oder ähnlicher Form keiner anderen Prüfungsbehörde vorgelegt und auch nicht veröffentlicht.

Esslingen, den 28. März 2026 ________________________

Unterschrift

Sperrvermerk

Die nachfolgende Master's Thesis enthält vertrauliche Daten der Fraunhofer IPA. Veröffentlichungen oder Vervielfältigungen dieser Arbeit – auch nur auszugsweise – sind ohne ausdrückliche Genehmigung der Fraunhofer IPA nicht gestattet. Diese Arbeit ist nur den Prüfern sowie den Mitgliedern des Prüfungsausschusses zugänglich zu machen.

Zitat

„Far out in the uncharted backwaters of the unfashionable end of the western spiral arm of the Galaxy lies a small unregarded yellow sun. Orbiting this at a distance of roughly ninety-two million miles is an utterly insignificant little blue green planet whose ape-descended life forms are so amazingly primitive that they still think digital watches are a pretty neat idea.“

Douglas Adams – The Hitchhikers Guide to the Galaxy

Vorwort

Dank an die Firma und die Firmenmitarbeiter, max. 1/2 Seite

Kurz-Zusammenfassung

„Aushängeschild"“ der Arbeit, max 1 Seite

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1 Milestone Timeline

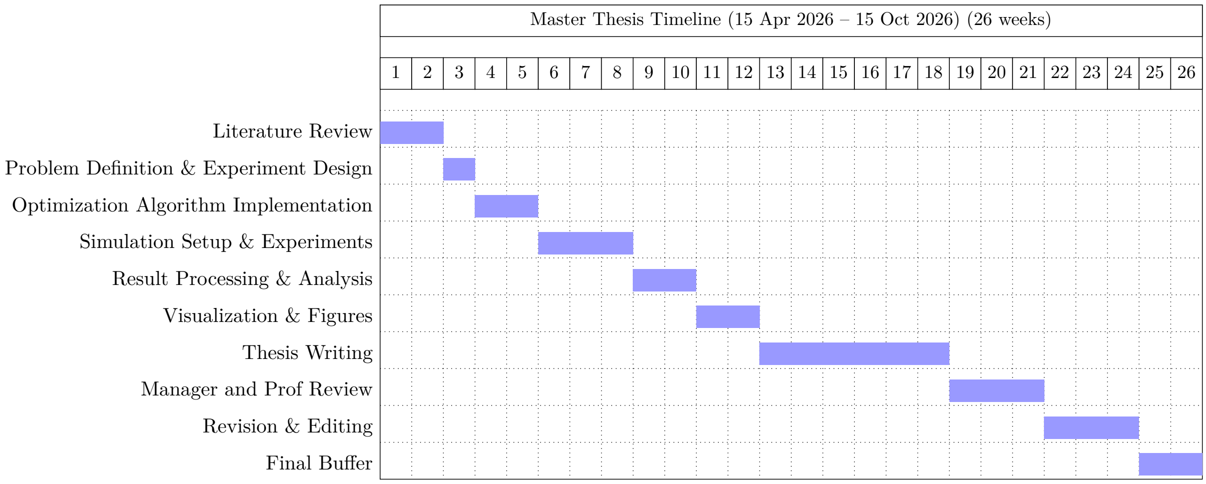

Abb. 1.1: Master thesis milestone plan

2 Einleitung

Die Gliederung des Hauptteils dieses Dokuments ist nur als Platzhalter gedacht. Im Folgenden sind zwei Vorschläge für Gliederungen dargestellt, die aber stets am konkreten Fall überprüft und in der Regel angepasst werden müssen.

Bitte beachten: Umfangreiche Programmlistings gehören nicht in die Arbeit, sondern auf eine beigefügte CDROM. Programmbeispiele können auszugsweise in der Arbeit gelistet werden, wenn sie zum Verständnis notwendig sind.

2.1 Literaturarbeit

1. Überblick (oder: Zusammenfassung, „Executive Summary", alles Wichtige für den „Manager" oder Schnellleser)
2. Fragestellung (oder: Ziele, Ausgangspunkt, Motivation)
3. Übersicht über den Stand der Wissenschaft und Technik (Beschreibung der Lösungsansätze, Beispiele etc. in einzelnen Abschnitten)
4. Bewertung der einzelnen untersuchten Ansätze, Beispiele etc., Identifikation von Defiziten
5. Synthese: Erstellung einer Gesamtschau, allgemeine Prinzipien, Beschreibung einer eigenen Sicht auf das Problem, evtl. auch eigene Vorschläge
6. Zusammenfassung (Erklärung des Nutzens), Ausblick

Anhang: eventuell recherchierte Texte, Produktbeschreibungen, etc.

2.2 System- und Softwareentwicklungen

1. Überblick (oder: Zusammenfassung, „Executive Summary", alles Wichtige für den „Manager" oder Schnellleser)
2. Problemstellung (oder: Ziele, Ausgangspunkt), Vorgesehener Benutzerkreis, Bedürfnisse der Benutzer
3. Stand der Technik (Wie wird das Problem bisher gelöst, wo sind die Defizite)
4. Gewählter Lösungsansatz (allgemeines Prinzip, welche Werkzeuge, z.B. Programmiersprachen werden verwendet)
5. Beschreibung der durchgeführten Arbeiten

6. Ergebnis (z.B. Screenshots mit Erläuterungen, Bedienanleitung)
7. Zusammenfassung (Erklärung des Nutzens), Ausblick

Anhang: evtl. (ausgewählte) Programmbeispiele.

3 Introduction

3.1 Motivation

Robotic grasping is a fundamental capability in automated manufacturing, logistics, and service robotics. The ability to reliably grasp objects of varying shapes, sizes, and surface properties remains a challenging problem, particularly in unstructured environments. Industrial applications such as bin picking, assembly, and material handling require grippers that can operate robustly across a wide range of workpieces.

Among the commonly used end-effectors, suction-cup grippers and two-finger grippers represent two complementary approaches. Suction grippers are widely used for handling objects with smooth surfaces due to their simplicity and speed, whereas two-finger grippers provide flexibility in handling objects with complex geometries. However, designing optimal configurations for these grippers is non-trivial, as grasp performance depends on multiple interacting factors such as geometry, contact distribution, force limits, and collision constraints.

In particular, multi-suction cup grippers introduce additional complexity due to the spatial arrangement of suction cups. The positioning of the suction cups directly influences grasp stability, force distribution, and robustness to disturbances. Similarly, for two-finger grippers, parameters such as finger geometry and actuator limits affect grasp feasibility and performance.

These challenges motivate the use of systematic optimization techniques to identify high-quality gripper configurations. Since multiple objectives must be considered simultaneously, such as maximizing stability while ensuring robustness and feasibility, the problem is naturally formulated as a multi-objective optimization problem.

3.2 Goal and Tasks of the Thesis

The primary goal of this thesis is to develop and evaluate a multi-objective optimization framework for determining optimal configurations of suction-cup and two-finger grippers.

To achieve this goal, the following tasks are defined:

- Formulation of the gripper configuration problem as a multi-objective optimization problem, including the definition of decision variables, objective functions, and constraints.
- Development of a evaluation pipeline for grasp generation, collision checking, and grasp quality assessment using point cloud representations of workpieces.
- Implementation of an optimization framework capable of handling multiple objectives and generating Pareto-optimal solutions.

- Application and comparison of different optimization algorithms, including Bayesian-based optimization methods, and Evolutionary Algorithms.
- Analysis of the influence of gripper parameters on grasp performance, with particular focus on multi-suction cup arrangements and two-finger gripper configurations.

3.3 Approach

To address the defined objectives, a multi-objective optimization framework is developed.

The gripper configurations are parameterized using a set of decision variables derived from configuration files. For the multi-suction cup gripper, the positions of the suction cups are defined in the planar space, forming a triangular arrangement. The height and orientation of the suction cups are assumed to be fixed. Each candidate solution is therefore represented by the coordinates of three suction cups. In this work, a triangular configuration of suction cups is assumed to yield a stable grasp. This work assumes that a triangular arrangement of suction cups leads to a stable grasp configuration. While alternative configurations may exist, the triangular layout provides a minimal and geometrically robust structure for distributing contact forces across the workpiece surface.

An initial population of candidate configurations is generated and iteratively refined using optimization algorithms. For each candidate solution, a subprocess is executed to evaluate grasp performance. In this subprocess, grasp candidates are generated for each point of the workpiece point cloud. A collision detection step ensures that only physically feasible grasps are retained by filtering out configurations where the gripper intersects with the workpiece.

Due to the potentially large number of valid grasp candidates, a clustering step is performed to group similar grasps and reduce redundancy. The number of clusters is treated as a hyperparameter. If the number of valid grasps is below this threshold, clustering is skipped.

Based on the filtered and clustered grasps, objective functions are computed. These include the maximization of the area of the triangular arrangement of suction cups, which is associated with grasp stability, and the maximization of a diversity metric that measures the spatial distribution of valid grasp clusters across the workpiece surface.

The evaluated objective values are then used by the optimization algorithms to generate new candidate solutions. This iterative process continues until a termination criterion is reached. The result of the optimization is a set of Pareto-optimal gripper configurations that represent trade-offs between competing objectives.

The same framework is extended to two-finger grippers by adapting the decision variables and objective functions to reflect their specific kinematic and geometric properties.

4 Methodology

4.1 Methodology for Multi-Suction Cup Gripper Optimization

This section describes the optimization framework developed for a multi-suction cup gripper consisting of three suction cups arranged in a triangular configuration.

4.1.1 Parameterization of the Gripper

Each suction cup is defined by its planar position, while the height and orientation remain fixed. Thus, the decision variables are:

$$\mathbf{p} = [(x_1, y_1), (x_2, y_2), (x_3, y_3)] \tag{4.1}$$

Each individual $\mathbf{p}$ forms a triangular base of the gripper.

Assumption In this work, a triangular configuration of suction cups is assumed to yield a stable grasp. This is motivated by the fact that three spatially distributed contact points form a statically stable support, enabling effective load distribution and resistance to external disturbances.

4.1.2 Initialization

An initial population of candidate solutions is randomly generated:

$$\mathcal{P} = \{\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_N\} \tag{4.2}$$

4.1.3 Workpiece Representation

Each workpiece is represented as a point cloud (PCD), where each point corresponds to a potential grasp location.

4.1.4 Optimization Loop

The optimization proceeds iteratively. For each individual $\mathbf{p} \in \mathcal{P}$, a subprocess is executed to evaluate grasp quality.

Subprocess: Grasp Evaluation

For each individual $\mathbf{p}$, the following steps are performed:

1. **Grasp Generation**

   For each point in the workpiece point cloud, a candidate suction grasp is generated by aligning the tool center point (TCP) of the gripper with the surface point.

2. **Collision Filtering**

   The gripper is modeled as a rigid 3D body with fixed height, forming a triangular prism whose base is defined by $\mathbf{p}$.

   Each candidate grasp is checked for collisions between the gripper body and the workpiece:

   - If a collision occurs, the grasp is discarded.
   - Otherwise, the grasp is considered valid.

   This step filters out physically infeasible grasps.

3. **Clustering of Valid Grasps**

   Let $\mathcal{G}_{valid}$ denote the set of valid grasps.

   A fixed number of clusters K is defined as a hyperparameter.

   - If $|\mathcal{G}_{valid}| <$ `nb_grasps_clustering`, clustering is skipped.
   - Otherwise, clustering is performed based on spatial proximity and orientation similarity.

   This step reduces redundancy and groups similar grasps.

4. **Triangle Area Calculation**

   The area of the triangle formed by the suction cups is computed as:

   $$A(\mathbf{p}) = \frac{1}{2} \left| (x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1) \right| \tag{4.3}$$

5. **Diversity Metric Calculation**

   A diversity score $D(\mathbf{p})$ is computed based on the spatial distribution of the clustered grasps:

   - Distances between cluster centers are calculated and normalized.
   - Orientation differences between grasps.

The diversity metric encourages wide coverage of the workpiece surface.

Objective Evaluation

After the subprocess, each individual $\mathbf{p}$ is assigned objective values:

$$\mathbf{f}(\mathbf{p}) = [-A(\mathbf{p}), -D(\mathbf{p})] \tag{4.4}$$

where:

- $A(\mathbf{p})$ is maximized (larger triangle improves stability)
- $D(\mathbf{p})$ is maximized (greater diversity improves robustness)

Population Update

The evaluated individuals are passed to a multi-objective optimization algorithm, which generates a new population:

$$\mathcal{P}^{(t+1)} = \text{Optimizer}(\mathcal{P}^{(t)}, \mathbf{f}) \tag{4.5}$$

The process is repeated until a termination criterion is satisfied.

4.1.5 Hyperparameters

The framework includes several predefined hyperparameters:

- Population size N
- Number of clusters `nb_grasps_clustering`
- Maximum number of iterations

These parameters influence the convergence behavior and computational cost of the optimization.

4.1.6 Output

The optimization produces a set of Pareto-optimal solutions:

$$\mathcal{P}^* = \{\mathbf{p} \mid \mathbf{p} \text{ is Pareto-optimal}\} \tag{4.6}$$

These solutions represent optimal trade-offs between triangle area and grasp diversity.

5 Stand der Technik

Beschreibung des momentanen Standes der Technik und Wissenschaft. Hier muss auch klar definiert werden, in welchem Zusammenhang das Problem steht, d.h. der Systemkontext und der bekannte Lï¿½sungsraum ist klar zu beschreiben (vorhandene.Lï¿½sungen

5.1 Problembeschreibung

5.2 Existierende Lï¿½sungen

5.3 Literaturverweise

Verweise im Text: [1] und [Gun04].

6 Lï¿½sungsansatz

Hier kï¿½nnen je nach Art der Arbeit auch verschiedene Lï¿½sungsansï¿½tze beschrieben und diskutiert werden.

7 Durchgefï¿½hrte Arbeiten

Es muss klar erkennbar sein, auf was aufgesetzt wurde, d.h. schon vorhanden war, und was im Rahmen dieser Arbeit selbst entwickelt wurde.

8 Ergebnisse

Hier kann z.B. eine Bedienanleitung stehen, oder bei Analysen die Analyseergebnisse. „Neuigkeiten“ Messergebnisse

9 Zusammenfassung

Kurze Zusammenfassung und evtl. Beschreibung der Punkte, die zu einer kompletten, marktreifen Lï¿½sung noch fehlen.

10 Schluss

Ergebnis-Bewertung, Zusammenfassung und Ausblick

A Kapitel im Anhang

Alles was den Hauptteil unnötig vergrößert hätte, z. B. HW-/SW-Dokumentationen, Bedienungsanleitungen, Code-Listings, Diagramme

Literaturverzeichnis

[1] Thomas Nonnenmacher, LaTeX Grundlagen - Setzen einer wissenschaftlichen Arbeit Skript, 2008,http://www.stz-softwaretechnik.de; *(Bei STZ Internetseite unter Publikationen - Skripte) [V. 2.0 26.02.08]*

[Gun04] Karsten Günther, LaTeX2 — Das umfassende Handbuch, Galileo Computing, 2004, http://www.galileocomputing.de/katalog/buecher/titel/gp/titelID-768; *1. Auflage*